

Predicting and validating a complex

the IFT52:IFT46 interaction, and how to read the PAE

Computer exercise (Part 2): AI for structure prediction

Cilia Connect 2026, University of Bonn

Aim of this exercise

In Part 1 you predicted the 3D structure of a single protein and learned to read the pLDDT and pTM confidence scores. A prediction becomes far more powerful when you ask it about a **complex**: do two proteins actually bind, and if so, how? Here you will predict the 3D structure of a real ciliary complex, the two intraflagellar-transport proteins **IFT52** and **IFT46**, and judge whether AlphaFold's predicted interaction can be trusted. You will meet a low overall score, work out for yourself whether a real interface hides behind it, and then test your idea by re-predicting a minimal complex. The key tools are the **PAE** plot and the **ipTM** score.

1 Submit the complex prediction

Task. On alphafoldserver.com, start a new job and add **two protein entities**: paste the IFT52 sequence into the first and the IFT46 sequence into the second. Name the job IFT52_46 and run it, then leave it computing.

These are the full-length *Chlamydomonas* sequences for IFT52 and IFT46. Working with the whole proteins is deliberate: part of the exercise is deciding for yourself which parts of them matter.

CrIFT52 (paste into entity 1)

```
MEEPGAEEVRILFSTAKGESHTHKAGFKQLFRRLRSTYRDPKVDKDDFTLDTLRSAILVLGGPKEKF  
TAPEVDMLKKFVKNGGSILILMSEGGEKAGTNINYFLEQFGMSVNNDAVVRTTHYKYLHPKEVLISD  
GILNRAVITGAGKSLNSDDDEFVSRGPQAFDGTGLEIVFPFGATLSVQKPAVPVLSSGKIAYPMNR  
PVGAVWAQPGYGRIVLGSAMFDDKWLDKEENSKIMDFFFKFLKPHSKIQLNDIDAEEDVSDLKLL  
PDTASLADKLKGCLQEIDDVPRDWTSLFDDSLFKFDTGLIPEAVSLYEKLGVKKGQLNLIPPSFETPL  
PPLQPAVFPPTIREPPPALLELFDLDESFASETNRLASLTNKCHGEEDLEYIMEAGHILGLKLQEN  
ANAKHVLSEVFRRIAQYKMGSLGLGQTLDSMGQTLPAANQFGDQFEL
```

CrIFT46 (paste into entity 2)

```
MDDSDYPDRDGGDLQFQGTARSQVVQNQPHDEEVNLSESESFAGADEPPAAPRDASLIESHMDDEG  
PAAPARTLSPTGYEAGKHAPGGIANSDEAPPGAYNAQEYKHLNVGEDVRELFYIGRYKPQTVELDTR  
IKPFIPDYIPAVGGIDEFIKVPDPDKPDYLGKVLDEPAKQSDPTVTLTLQLRQLSKEAPGAKADMV  
GRLEHTDENAKKIQWIASINDIHKAKPAATVNYSKRMPEIEALMQEWPPEVETFLKTMHMPSGDVE  
LDIKTYARLVCTLLDIPVYDDPVESLHVLFYLYLEFKNNPIFRQHMEMENKLDGMSGGGGMMGGGAD  
VLGL
```

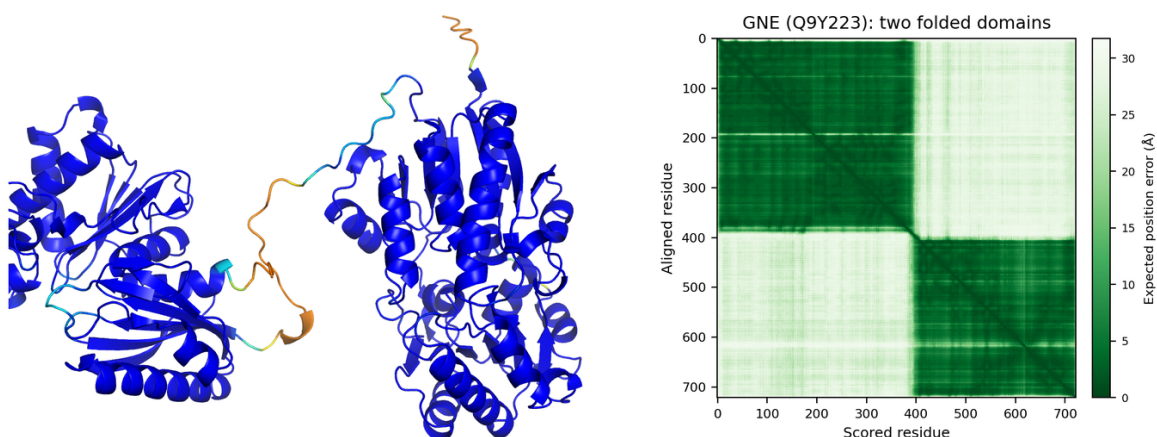
Note. Once everyone has submitted, we pause here for a short presentation on how to judge a predicted complex: the **PAE** plot and the **ipTM** score. After that, keep working through the sections below while your job runs.

2 While it runs: practise reading a PAE plot

You have just seen what the **Predicted Aligned Error (PAE)** reports: for a pair of residues, how sure AlphaFold is about their position *relative to each other* (dark = confident, light = not). AlphaFold has already predicted the structure of over 214 million proteins, essentially every sequence in the UniProt database, and more recently almost 80,000 high-confidence protein complexes; you can look them up on the EBI AlphaFold database. Practise reading the PAE on two of them before you turn to your own complex.

2.1 Two folded domains that do NOT interact

Task. Open the AlphaFold database entry for the human protein **GNE** (accession Q9Y223, entry AF-Q9Y223-F1), and find the **Predicted Aligned Error** viewer. The database also lists a predicted GNE *homodimer* (AF-0000000065914703), whose PAE has two chains and looks quite different: make sure you are on the single-chain entry. The plot is reproduced below so you can also do this offline.

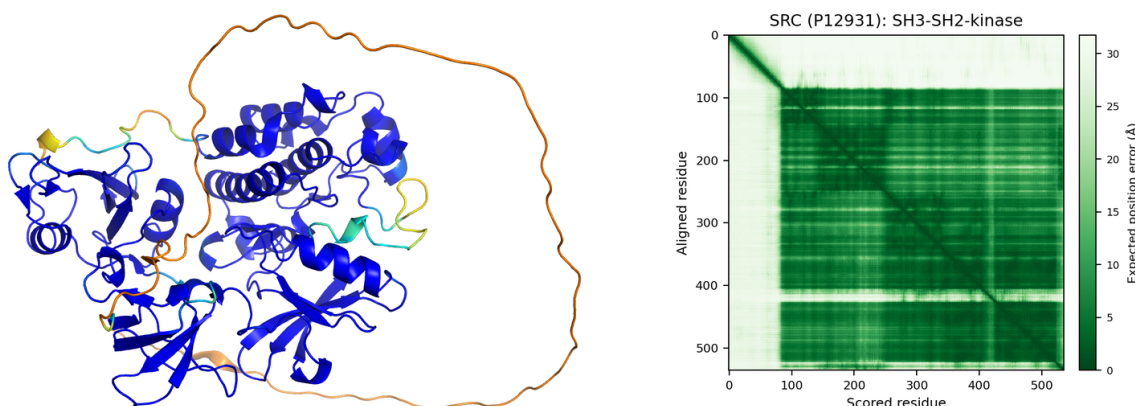


GNE has two well-folded domains (each pLDDT > 90, blue) joined by a flexible linker (orange). The structure is coloured by pLDDT, the PAE is on the right.

Question. Aligning on a residue in the first domain (~300), are you confident about another residue in the *same* domain (~200)? About a residue in the *other* domain (~600)? Each domain scores high pLDDT on its own, so what is the off-diagonal block telling you that pLDDT alone did not? Do the two domains have a fixed position relative to each other?

2.2 A protein whose parts ARE locked together

Task. Now look up the human tyrosine kinase **SRC** (accession P12931). Its PAE is shown below.



SRC has three folded domains, SH3 (residues 84–145), SH2 (151–248) and a kinase domain (~267–520), shown in blue, preceded by a long disordered N-terminal region (~1–83, orange). Structure coloured by pLDDT on the left, PAE on the right.

Question. Read this plot the same way. Compared with GNE, what is different about the large off-diagonal region among the folded domains? Which residues, by contrast, stay light against everything else, and what does that tell you about that part of the protein? In one sentence: which parts of SRC are held in a fixed position relative to each other, and which are not?

The step to carry over to your complex. Two domains of one protein and two chains of a complex read the same way. In a two-protein job, look at the off-diagonal block *between the chains*: **dark** = a real, confident interface (like SRC's folded core); **light** = the parts float apart (like GNE's two domains). The single number that summarises it is the **ipTM** (>0.8 confident, 0.6–0.8 borderline, <0.6 do not trust).

3 Examine your complex

3.1 Open the top model and its PAE

Task. Download the IFT52_46 results and unpack the zip. Open the top-ranked model (the `...model_0.cif` file) in PyMOL, and on the AF3 server open the **PAE plot** for the same job.

Question. Is the whole inter-chain region of the PAE light, or is there a *compact dark block* somewhere in it? If there is a block, which stretch of each chain does it cover? Putting the plot together with the structure, do the two proteins interact, and if so, which parts of them?

3.2 Now read the ipTM

Question. Finally, read the **ipTM** for the top model. Is it high and confident? If not, why not, given what you have just seen? The ipTM is a single average over *all* residue pairs between the two chains: if only a small part of each protein forms the interface and the rest does not, how does that pull the average down? Does a low ipTM here mean the two proteins do not interact?

4 Which residues form the interface?

You have found that the two proteins touch in a specific region. Now pin down exactly which residues. In the top model (chain A = IFT52, chain B = IFT46), let PyMOL pick the contacting residues:

```
select ifaceA, byres (chain A within 5 of chain B)
select ifaceB, byres (chain B within 5 of chain A)
iterate ifaceA and name CA, print("IFT52", resi)
iterate ifaceB and name CA, print("IFT46", resi)
```

Now show the interface residues as sticks, colour the two chains differently, and let PyMOL draw the polar contacts (hydrogen bonds and salt bridges) between them as yellow dashes:

```
show sticks, (ifaceA or ifaceB) and not name N+C+O
color skyblue, chain A and elem C
color lightorange, chain B and elem C
distance polar, ifaceA, ifaceB, mode=2
hide labels, polar
zoom ifaceA or ifaceB
```

Question. First, read off the residue ranges for each chain. Do they match the stretches of IFT52 and IFT46 that formed the dark inter-chain block in the PAE plot?

Question. Second, examine the interface in detail. What holds the two proteins together: is it mainly hydrophobic side chains packing against each other, hydrogen bonds, salt bridges between charged residues, or a mix? Find an example of each kind you see, and note which residues are involved.

5 Test it: predict a minimal complex

You now have a hypothesis: the interaction lives in the region you just found, and the rest of each protein is only diluting the score. Test it.

Task. Submit a new AF3 job containing *only* the interacting parts you identified (the residue ranges of IFT52 and of IFT46 that make the contacts). Name it IFT52_46_trunc.

Question. Before it finishes, **predict:** what will happen to the ipTM now that the non-interacting parts are gone, and what will happen to the *shape* of the PAE plot? When the job returns, compare it with the full-length result. Were you right? What does the change tell you about what the ipTM was actually measuring before?

Note. The truncated prediction takes a while to come back. While it runs you do not need to wait: read the final section (a word of caution), and, if you like, try the optional interface analysis below. Return here to compare the two results once the truncated job finishes.

6 A prediction still needs validation

Everything so far builds confidence: you found a compact interface, the PAE backed it up, and truncating to it pushed the ipTM from low to confident. But all of this is still a **prediction**. A convincing PAE and a good ipTM make a strong hypothesis, not a proven fact. Before you build on a predicted interface, it has to be tested experimentally: treat every AlphaFold complex as a lead to be confirmed, not as the answer.

This particular interface happens to have been confirmed: the C-terminal domains of IFT52 and IFT46 were solved by X-ray crystallography (PDB 4UZZ), so here you can check your prediction directly.

Task. *If time permits.* In PyMOL, fetch the experimental structure and superpose your complex onto it (your truncated model is cleanest):

```
fetch 4uzz
super <your model>, 4uzz
```

Question. Does your predicted interface land on the experimental one, and what RMSD do you get? The crystal structure is from a different organism, so expect a small offset of a couple of Å rather than zero.

Take-home messages: pLDDT is local; the PAE and ipTM describe relative position and interfaces; ipTM is a *global* average, so a real but localized interface can hide behind a low ipTM: read the PAE, find the block, and test it by re-predicting a minimal complex; and even a convincing prediction stays a hypothesis until it is confirmed experimentally.

Optional: read the interface straight from the numbers. You just picked the interface residues off the structure. The same interface is already in the PAE, as numbers. Write a short Python script that opens the `*_full_data_*.json` file, reads the PAE matrix, and lists the *inter-chain* residue pairs whose PAE is below about 5 Å (use the `token_chain_ids` entry in the file to tell the two chains apart). Do those low-PAE pairs fall in the same region you found by contacts in PyMOL? If you are comfortable with Python you can do this yourself; otherwise a coding assistant (an LLM) will get you there quickly.

Optional genius task: build a better ipTM. You have seen that the ipTM undersells a real but small interface between two large proteins. Now that you can read the PAE matrix from the json file, try to fix that.

How AlphaFold computes the ipTM, in short: for a residue i in one chain, it turns the PAE to every residue j in the *other* chain into a score between 0 and 1,

$$f_{ij} = \frac{1}{1 + (\text{PAE}_{ij}/d_0)^2}, \quad d_0 = 1.24(N - 15)^{1/3} - 1.8,$$

and averages f_{ij} over all those j . The ipTM is the highest such average over all residues i . N is the total number of residues in the complex.

One thing in this recipe punishes a small interface on large proteins: find it and fix it. Then ask whether d_0 , which depends on N , still makes sense for the residues you are now scoring, and adjust it. Write your own score. Test it on both your full-length and your truncated prediction: does your score give similar values for the two, where the ipTM did not?

See Dunbrack, R.L. (2025). *Rēs ipSAE loquunt: What's wrong with AlphaFold's ipTM score and how to fix it.* *bioRxiv*, doi:10.1101/2025.02.10.637595, for help.